Chapter-3

Numerical Differenciation

Let, $f(x) = 2x^2 - e^x$

Using this function approximate $f'(2)$

Using central difference:

h	$f'(x_0)$	Truncation Error
0.1	0.55860	0.01234
0.01	0.61050	4.4×10^{-4}
0.001		

central difference, $f'(x) = \dfrac{f(x_0+h) - f(x_0-h)}{2h}$

So, for h_0, $f'(x) = \dfrac{f(2+0.1) - f(2-0.1)}{2(0.1)}$

$= \dfrac{0.65803 - 0.53411}{2 \times 0.1}$

$\dfrac{f(2+0.01) - f(2-0.01)}{2(0.01)}$

Forward Difference ; $f'(x)_{approx} = \frac{f(x_0+h)-f(x_0)}{h}$

Backward " ; $f'(x)_{approx} = \frac{f(x_0)-f(x_0-h)}{h}$

Central " ; $f'(x)_{approx} = \frac{f(x_0+h)-f(x_0-h)}{2h}$

$f(x) = 2x^2 - e^x$

$f'(x) = 4x - e^x$

So, $f'(2) = 4(2) - e^2 = 0.61054$ [Accurate]

Truncation Error = Accurate val - Approx Val

$= 0.61054 - 0.59860$

$= 0.01234$

0.1

Here,



$x_0 = 4.0, 4.1, 4.2, 4.3, 4.4$

$f(x_0) = 16, 18, 20, 21, 22$

Using F.D calculate the value of $f'(4.2)$.

and BD;

FD

Here, $h = 0.1$

$$f'(4.2) = \frac{f(4.2+0.1) - f(4.2)}{0.1}$$

$$= \frac{21 - 20}{0.1} = 10$$

BD

$$f'(4.2) = \frac{f(4.2) - f(4.2-0.1)}{0.1}$$

$$= \frac{20 - 18}{0.1}$$

$$= 20 \quad (\text{Ans.})$$

$v(t) = 2000 \ln \left[\frac{14 \times 10^4}{14 \times 10^4 - 21nt} \right] - 9.8t$

$v'(16) = ?$ $\Delta t = 2$

$$FD = \frac{v(18) - v(16)}{2}$$

$$BD = \frac{v(16) - v(14)}{2}$$

$$CD = \frac{v(18) - v(14)}{2 \times 2}$$

Forward diff

$f(x) = \ln(x)$

Here, $f(x) = \ln(x)$

So, $f'(x) = \frac{1}{x}$

$f'(2) = \frac{1}{2} = 0.5$

Using forward Differentiation, $\frac{f(x_0+h) - f(x_0)}{h}$

h	$f'(x_0)$	Truncation Error
	$\frac{f(x_0+h) - f(x_0)}{h}$	[Actual val − Forward diff val]
	$\rightarrow \frac{\ln(2+h) - \ln(2)}{2}$	$\rightarrow$ [0.5 − Forward diff val]
1	0.405465	0.0945345
0.1	0.487902	0.0120984
0.01	0.498754	0.00124585
0.001	0.499875	0.00012

Decreasing on a scale of 10, just like h

if h is divided by 10, error also gets divided by 10.

$\therefore$ error $\propto h$

Backward Diff

Error $\propto h$ [Derivation same as F.D]

Central Diff

$$\text{Error} \propto h^2 \left[\frac{f(x+h) - f(x-h)}{2h} - \frac{f'''(\xi)h^2}{3!} \right]$$

For CD, Error becomes small when $h < 1$

$f(x) = \ln(x)$ $\quad f'(x) = \frac{1}{x}$; $f'(2) = \frac{1}{2} = 0.5$

h	$f(x_0)$	Truncation Error
	$\frac{f(x_0+h) - f(x_0-h)}{2h}$	$\|$Actual val − cd val$\|$
1	0.549306	0.049306
0.1	0.500417	0.000417293
0.01	0.500004	4.16673×10^{-6}
0.001	0.500000	4.16666×10^{-8}

If h is divided by 10, error gets divided by 100.

$\therefore$ error $\propto h^2$

Rounding Error

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h}$$

(i) Smaller step size h, better result.

(ii) However, if h is very small, $f(x+h)$ and $f(x-h)$ will have similar values.

Ex → $f(2+0.001) \approx f(2-0.001)$

(iii) In floating point chapter, we learned about loss of significance.

(iv) When we subtract 2 numbers which are close to each other, there are large errors.

(v) When h tends to 0 (very small), $f(x+h)$ & $f(x-h)$ are numbers which are closer to each other.

(vi) Therefore $\frac{f(x+h) - f(x-h)}{2h}$ will give large errors (large rounding error)!

From chapter-1, we know,

$$fl(x) = (1+\delta_1)\, x$$

So, $fl[f(x_1+h)] = (1+\delta_1)\, f(x_1+h)$

$fl[f(x_1-h)] = (1+\delta_2)\, f(x_1-h)$

$|\delta_1|, |\delta_2| \leq \phi \epsilon_m$

Calculating the error,

$$\left| f'(x_1) - \frac{fl[f(x_1+h)] - fl[f(x_1-h)]}{2h} \right|$$

|Actual value of differentiation − value of differentiation by numerical approach|

$$= \left| \frac{f(x_1+h) - f(x_1-h)}{2h} - \frac{f'''(\xi)}{3!} h^2 - \frac{fl[f(x+h)] - fl[f(x_1-h)]}{2h} \right|$$

$$= \left| \frac{f(x_1+h) - f(x_1-h)}{2h} - \frac{f'''(\xi)}{3!} h^2 - \frac{(1+\delta_1) f(x_1+h) - (1+\delta_2) f(x_1-h)}{2h} \right|$$

$$= \left| -\frac{f'''(\xi)}{6} h^2 - \frac{\delta_1 f(x_1+h) - \delta_2 f(x_1-h)}{2h} \right|$$

$|a+b| \leq |a| + |b| \rightarrow |5+(-1)| \leq |5| + |-1|$

$|4| \leq 5+1$

$|4| \leq 6.$

$$\leq \left| \frac{f'''(\xi)}{6} \right| h^2 + \frac{|\delta_1 f(x_1+h) - \delta_2 \cdot f(x_1-h)|}{2h}$$

Now, $|\delta_1|, |\delta_2| \leq \epsilon_m$

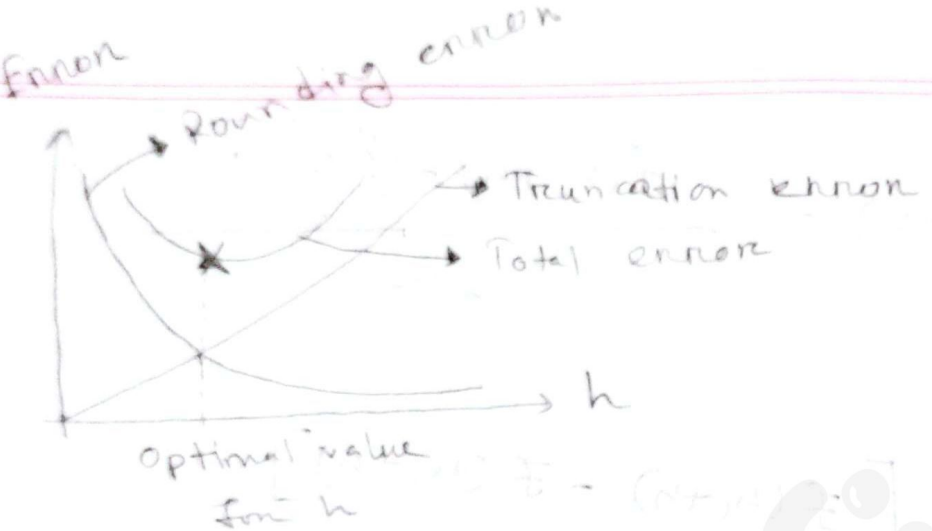
$$\leq \underbrace{\frac{|f'''(\xi)|}{6} h^2}_{\text{Truncation Error}} + \underbrace{\epsilon_m \frac{|f(x_1+h)| + |f(x_1-h)|}{2h}}_{\text{Rounding Error}}$$

Truncation Error

- This term comes from the truncation part of the series
- Smaller h, lesser error.

Rounding Error

- Smaller h, large error
- Since h is in the denominator.



Richardson Extrapolation

We know, Taylor Series,

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''}{2!}(x_1)(x-x_1)^2 + \cdots$$

$D_h = CD \rightarrow f'(x_1) = \dfrac{f(x_1+h) - f(x_1-h)}{2h}$

Now,

$$f(x_1+h) = f(x_1) + f'(x_1)h + \frac{f''(x_1)}{2!}h^2 + \frac{f^3(x_1)}{3!}h^3 + \frac{f^4(x_1)}{4!}h^4 + \frac{f^5(x_1)}{5!}h^5 + O(h^6)$$

$$f(x_1-h) = f(x_1) - f'(x_1)h + \frac{f''(x_1)}{2!}h^2 - \frac{f^3(x_1)}{3!}h^3 + \frac{f^4(x_1)}{4!}h^4 - \frac{f^5(x_1)}{5!}h^5 + O(h^6)$$

As,

$$D_h = f'(x_1) = \frac{f(x_1+h) - f(x_1-h)}{2h}$$

Now,

$$D_h = \frac{1}{2h}\left[f(x_1+h) - f(x_1-h)\right]$$

$$= \frac{1}{2h}\left[2f'(x_1)h + \frac{2f^3(x_1)}{3!}h^3 + 2\frac{f^5(x_1)}{5!}h^5 + O(h^7)\right]$$

যেহেতু $O(h^6)$ কাছাকাছি চলে আসে, তাই পরের পদ $O(h^7)$ আকার

$$= \underbrace{f'(x_1)}_{\text{Exact Val).}} + \underbrace{\frac{f^3(x_1)}{3!}h^2 + \frac{f^5(x_1)}{5!}h^4 + O(h^6)}_{\text{Error}}$$

Exact value $\longrightarrow f'(x_1)$

Error $\longrightarrow \frac{f^3(x_1)}{3!}h^2 + \frac{f^5(x_1)}{5!}h^4 + O(h^6)$

Dominating factor of the Error $\longrightarrow \frac{f^3(x_1)}{3!}h^2$

We need to reduce it from the series ↑

That's why we introduce $D_{h/2}$

→ Here, Error is of order h^2, because $h^4, h^6 \ldots$ are less dominant than h^2.

→ Hence proving again that, error ~ h^2 for CD

So, we need to take combinition in such way that h^2 term goes way. So that, we are left with h^4.

$$D_h = f'(x_1) + \frac{f^3(x_1)\,h^2}{3!} + \frac{f^5(x_1)}{5!}h^4 + O(h^6)$$

$$D_{h/2} = f'(x_1) + \frac{f^3(x_1)}{3!}\left(\frac{h}{2}\right)^2 + \frac{f^5(x_1)}{5!}\left(\frac{h}{2}\right)^4 + O(h^6)$$

$$2^2 D_{h/2} = 2^2 f'(x_1) + \frac{f^3(x_1)}{3!}h^2 + \frac{1}{2^2}\frac{f^5(x_1)}{5!}h^4 + O(h^6)$$

Now, $2^2 D_{h/2} - D_h$)

~~$2^2 D_{\frac{h}{2}} - D_h = 2^2 f'(x_1) - f'(x_1) + \ldots$~~

$$2^2 D_{\frac{h}{2}} - D_h = (2^2-1) f'(x_1) + \left(\frac{1}{2^2}-1\right)\left(\frac{f^5(x_1)}{5!}h^4\right) + O(h^6)$$

We need only $f'(x_1)$; so we need to get rid of (2^2-1) part. So, we need to divide both side with (2^2-1)

Now,

$$\frac{2^v D_{\frac{h}{2}} - D_h}{2^v - 1} = f'(x_1) + \frac{\left(\frac{1}{2^v} - 1\right)}{(2^v - 1)5!} f^5(x_1) h^4 + O(h^5)$$

$$D_h' = \frac{2^v D_{h/2} - D_h}{2^v - 1}$$

Now, if we take this combination, error gets reduced to an order of h.

Lets consider, $\frac{2^v D_{h/2} - D_h}{2^v - 1} = D_h'$

Now calculate, $D_{\frac{h}{2}}'$

Now, take combination in such a way that h^4 goes away. So, now we can have an error of order h^6. It will be called $D_h^{(2)}$ then.

So, $D_h = f'(x_1) + C h^n + O(h^{n+1})$

↳ C could be anything like $\frac{f^n(x_1)}{n!}$

$$D_{h/2} = f'(x_1) + C\left(\frac{h}{2}\right)^n + O(h^{n+1})$$

Multiply $D_{\frac{h}{2}}$ with 2^n; we get.

$$D_h^{(n)} = \frac{2^n D_{\frac{h}{2}} - D_h}{2^n - 1}$$

কিছু- এর প্রয়োগ এর প্রয়োগ use CD.

$f(x) = e^x \sin(x)$; Find $f''(1)$ using Richardson Extrapolation where, $h = 0.5$ and 0.25.

We know,

$$CD, f'(x) = \frac{f(x_0+h) - f(x_0-h)}{2h}$$

for $h = 0.5$

$$f(1+0.5) = e^{1.5} \cdot \sin(1.5) = 4.470$$

$$f(1-0.5) = e^{0.5} \cdot \sin(0.5) = 0.790$$

$$\text{So, } f'(1) = \frac{f(1+0.5) - f(1-0.5)}{2(0.5)}$$

$$= \frac{4.470 - 0.790}{1}$$

$$= 3.68$$

For $h = 0.25$

$$f(1+0.25) = e^{1.25} \cdot \sin(1.25) = 3.312$$

$$f(1-0.25) = e^{0.75} \cdot \sin(0.75) = 1.443$$

$$\text{So, } f'(1) = \frac{f(1+0.25) - f(1-0.25)}{2(0.25)}$$

$$= \frac{3.312 - 1.443}{0.5} = 3.738$$

Now,

step	Dn
h = 0.5	3.68
h = 0.25	3.738

So,

$$D'_h = \frac{2^r D\left(\frac{h}{2}\right) - D(h)}{2^r - 1}$$

$$= \frac{4 D(h/2) - D(h)}{3}$$

$$= \frac{4(3.738) - 3.68}{3}$$

$$= 3.757$$

(Ans.)

Find $f'(1)$ using Richardson Extrapolation.

Step(h)	$x[f'(1)]$
0.1	0.7
0.2	0.5

We know,

$$D_h^n = \frac{2^n D\left(\frac{h}{2}\right) - D_h}{2^n - 1}$$

So, $$D_h^1 = \frac{2^1 D\left(\frac{h}{2}\right) - D(h)}{2^1 - 1}$$

$$= \frac{4(0.7) - 0.5}{3}$$

$$= 0.767 \quad \text{(Ans.)}$$

$f(x) = e^{2x} + 3x$; Find $f'(2)$; $h = 1.2$ & 0.6

For $h = 1.2$

$$f'(2) = \frac{f(x_0+h) - f(x_0-h)}{2h} = \frac{f(2+1.2) - f(2-1.2)}{2(1.2)}$$

611.445

$$= \frac{\text{[illegible]} - 7.353}{2(1.2)}$$

$$= \text{[illegible]}\ 251.705$$

For $h=0.6$

$$f'(2) = \frac{f(2+0.6) - f(2-0.6)}{2(0.6)} = \frac{189.072 - 20.645}{1.2}$$

$$= 140.356$$

Now

$h=1.2;\ f'(2) = 251.705$

$h=0.6;\ f'(2) = 140.356$

$$D_n' = \frac{2^r D\left(\frac{n}{2}\right) - D(n)}{2^r - 1}$$

$$= \frac{4(140.356) - 251.705}{3}$$

$$= 103.239$$

(Ans.)

2.569347 ; 45F → 2.569
4DB → 2.5693

77

x	f(x)
0.6	0.707178
0.8	0.8553892
0.9	0.925863
1.0	0.984007
1.1	1.033243
1.2	1.074575
1.4	1.127986

~~Find f(0.4) for~~

Find $f'(1)$ for step h, 0.4, 0.2, 0.1

For 0.4

$$f'(1) = \frac{f(1+0.4) - f(1-0.4)}{2(0.4)} = \frac{f(1.4) - f(0.6)}{2(0.4)}$$

$$= \frac{1.127988 - 0.707178}{0.8}$$

$$= 0.52601$$

for 0.2

$$f'(1) = \frac{f(1+0.2) - f(1-0.4)}{2(0.2)} = 0.5465$$

for 0.1

$$f'(1) = \frac{f(1+0.1) - f(1-0.1)}{2(0.1)} = 0.5354$$

Now,

h	Dh
0.4	0.52601
0.2	0.5465
0.1	0.5394

৩টি দাশলে Highest 3 val নিচ, also Newton Method ✓ দিলাম, error না দিলে.

For 0.4 & 0.2

$$D'h = \frac{2^r D(\frac{h}{2}) - D(h)}{2^r - 1} = \frac{4(0.5465) - 0.52601}{3}$$

$$= 0.553$$

for 0.2 & 0.1

$$D'h = \frac{2^r D(\frac{h}{2}) - D(h)}{2^r - 1} = \frac{4(0.5394) - 0.5465}{3}$$

$$= 0.537$$

So, Now 0.4 & 0.2 & 0.1

$$D''h = \frac{2^4 D'(\frac{h}{2}) - D'(h)}{2^4 - 1} = \frac{16(0.537) - (0.553)}{15}$$

$$= 0.5355 \quad \text{(ans.)}$$

Upper Bound of Truncation Error in Differenciation

step size, $h = 0.2$

x	1.0	1.2	1.4
$f(x)$	1.2717	1.3642	1.5496

- Find $f'(1.2)$ using CD.

$$\text{So, } f'(1.2) = \frac{f'(1.2+0.2) - f'(1.2-0.2)}{2(0.2)}$$

$$= \frac{1.5496 - 1.2717}{0.4}$$

$$= 0.69475$$

- Compute the Truncation Error / Error bound using $\xi \to [1.0, 1.4]$ where, $f(x) = x\sin(x) + x\cos x$

$$\left| f^3(\xi) \frac{h^2}{3!} \right|$$

যেই মান বসালে $f(x)$ এর expression এর সবচেয়ে বড় মান পাব।

Here,

$$f(x) = x\sin x + x^2\cos x$$

$$f'(x) = \sin x + x\cos x + 2x\cos x - x^2\sin x$$

$$= \sin x + 3x\cos x - x^2\sin x$$

$$f''(x) = \cos x + 3\cos x - 3x\sin x - 2x\sin x - x^2\cos x$$

$$= 4\cos x - 5x\sin x - x^2\cos x$$

$$f'''(x) = -4\sin x - 5\sin x - 5x\cos x - 2x\cos x + x^2\sin x$$

$$= -9\sin x - 7x\cos x + x^2\sin x$$

Now, $\left|\frac{f^3(\xi)}{3!}h^3\right| = \left|9\sin(\xi) + 7(\xi)\cos(\xi) + \xi^2\sin(\xi)\right| \times \left|\frac{0.2^3}{6}\right|$

Upper Bound এর জন্য ξ এর যে value তে (1.4) এর আশে পাশে $\sin$ এর fm $\sin$ আর $\cos$ এর fm $\cos$ (1.0) এর আশে পাশে

$$= \left|9\sin(1.4) + 7(1.4)\cos(1.0) + (1.4)^2\sin(1.4)\right| \times \left|\frac{0.2^3}{3}\right|$$

$$= 16.096 \times \frac{1}{150} = 0.1073$$

Truncation Error = |Actual Val − Calculated val|.

Actual Error = Accurate Value − Approx Value

Accurate Value →

$$f(x) = x\sin(x) + x^2\cos x$$

So, $f'(x) = \sin(x) + 3x\cos(x) - x^2\sin x$

$$f'(1.2) = \sin(1.2) + 3(1.2)\cos(1.2) - (1.2)^2\sin(1.2)$$

$$= 0.8944$$

Approx Error →

CD পদ্ধতি ব্যবহার করে,

$$f'(1.2) = 0.69475$$

So, Actual Error = $0.8944 - 0.69475$

$$= 0.19965$$

(Ans.)

$$\text{Relative Error} = \left| \frac{\text{Truncation Error}}{\text{Actual Error}} \right|$$